
USACE / NAVFAC / AFCEC UFGS-23 65 00 (May 2025)

Preparing Activity: NAVFAC

Superseding
UFGS-23 65 00 (November 2016)

UNIFIED FACILITIES GUIDE SPECIFICATIONS

References are in agreement with UMRL dated January 2026

SECTION TABLE OF CONTENTS

DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)

SECTION 23 65 00

COOLING TOWERS AND REMOTE EVAPORATIVELY-COOLED CONDENSERS

05/25

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SAFETY REQUIREMENTS
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 PROJECT/SITE CONDITIONS
 - 1.5.1 Drawings

PART 2 PRODUCTS

- 2.1 STANDARD COMMERCIAL PRODUCTS
- 2.2 MANUFACTURER'S STANDARD NAMEPLATES
- 2.3 ELECTRICAL WORK
- 2.4 COOLING TOWER MATERIALS
 - 2.4.1 Fiberglass Reinforced Plastic (FRP)
 - 2.4.2 Zinc-Coated Steel
 - 2.4.3 Polyvinyl Chloride (PVC) Formed Sheets
 - 2.4.4 High Density Polyethylene (HDPE)
 - 2.4.5 Stainless Steel Sheets
 - 2.4.6 Concrete
 - 2.4.7 Hardware
- 2.5 OPEN CELL COOLING TOWERS
 - 2.5.1 Factory Assembled Towers
 - 2.5.1.1 Description
 - 2.5.1.2 Construction
 - 2.5.1.3 Tower Frame and Louvers
 - 2.5.1.4 Air Inlet and Discharge Connections
 - 2.5.1.5 Fill
 - 2.5.1.6 Drift Eliminators
 - 2.5.1.7 Cold Water Basin Equipment.
 - 2.5.1.7.1 Electric Basin Heater

- 2.5.1.8 Fans, Blowers, and Drives.
- 2.5.1.9 Tower Piping
- 2.5.1.10 Electric Motors
- 2.5.1.11 Vibration Cutout Switch.
- 2.5.1.12 Performance
- 2.5.1.13 Sound Power Level
- 2.5.1.14 Drift Loss
- 2.5.2 Lubrication
- 2.5.3 Factory Finish System
- 2.5.4 Field-Assembled Cooling Towers
 - 2.5.4.1 Framework, Casing, and Supports
 - 2.5.4.2 Foundations
 - 2.5.4.3 Stairways and Ladders
 - 2.5.4.4 Hand Railings
 - 2.5.4.5 Access Doors
 - 2.5.4.6 Louvers
 - 2.5.4.7 Fan Deck and Cylinder
 - 2.5.4.8 Fans
 - 2.5.4.9 Speed Reducers Gears and Drive Shaft
 - 2.5.4.10 Electric Motors
 - 2.5.4.11 Cold Water Basin
 - 2.5.4.12 Electric Basin Heater
 - 2.5.4.13 Hot Water Distribution System
- 2.5.5 Drift Eliminators
- 2.5.6 Cold Water Basin Equipment.
- 2.5.7 Fill (Heat Transfer Surface)
- 2.5.8 Fire Safety
- 2.5.9 Meters and Controls
- 2.6 CLOSED CIRCUIT COOLERS
 - 2.6.1 Description
 - 2.6.2 Casing and Finish
 - 2.6.3 Louvers
 - 2.6.4 Discharge Hood with Positive Closure Dampers
 - 2.6.5 Heat Exchanger
 - 2.6.6 Drift Eliminators
 - 2.6.7 Cold Water Basin and Intermediate Basins
 - 2.6.7.1 Electric Basin Heater
 - 2.6.8 Fans and Drives
 - 2.6.9 Piping
 - 2.6.10 Electric Motors
 - 2.6.11 Water Recirculation Pump
 - 2.6.12 Drift Loss
 - 2.6.13 Sound Power Level
 - 2.6.14 Vibration Cutout Switch
- 2.7 REMOTE EVAPORATIVELY-COOLED CONDENSERS
 - 2.7.1 Condenser Casing
 - 2.7.2 Refrigerant Section
 - 2.7.3 Fans
 - 2.7.4 Water Section
- 2.8 FABRICATION
- 2.9 SUPPLEMENTAL COMPONENTS/SERVICES
 - 2.9.1 Condenser Water Piping and Accessories
 - 2.9.2 Cooling Tower Water Treatment Systems
 - 2.9.3 Cooling Tower Passivation
 - 2.9.4 Cold Water Basin Sweeping System
 - 2.9.4.1 Basin Sweeping System
 - 2.9.4.2 Separator
 - 2.9.4.3 Filtration
 - 2.9.4.4 Pump Package

- 2.9.4.5 Power and Control
- 2.9.5 Temperature Controls

PART 3 EXECUTION

- 3.1 DEMONSTRATIONS
- 3.2 INSTALLATION
 - 3.2.1 Installation Instructions
 - 3.2.2 Vibration Isolation
 - 3.2.3 Posted Instructions
 - 3.2.4 Verification of Dimensions
 - 3.2.5 Demonstrations
 - 3.2.6 Certificates
 - 3.2.7 Start-Up Testing Report
 - 3.2.8 Operation and Maintenance Manuals
 - 3.2.9 Connections to Existing Systems
- 3.3 RELATED FIELD TESTING
 - 3.3.1 Test Plans
- 3.4 TESTING

-- End of Section Table of Contents --

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SECTION 23 65 00

COOLING TOWERS AND REMOTE EVAPORATIVELY-COOLED CONDENSERS 05/25

NOTE: This guide specification covers the requirements for induced mechanical draft cooling towers (both packaged and field-erected).

Adhere to [UFC 1-300-02](#) Unified Facilities Guide Specifications (UFGS) Format Standard when editing this guide specification or preparing new project specification sections. Edit this guide specification for project specific requirements by adding, deleting, or revising text. For bracketed items, choose applicable item(s) or insert appropriate information.

Remove information and requirements not required in respective project, whether or not brackets are present.

Comments, suggestions and recommended changes for this guide specification are welcome and should be submitted as a [Criteria Change Request \(CCR\)](#).

NOTE: Show the following information on the project drawings:

1. Design wet bulb temperature for capacity
2. GPM
3. Entering water temperature
4. Leaving water temperature
5. Motor voltage, phase, and horsepower of fans
6. Fan quantity
7. Sound rating.
8. Operating weight
9. Make-up water gpm
10. Vibration isolator type
11. When applicable, basin heater KW and voltage

- and phase.
- 12. Vibration isolation
- 13. When applicable, indicate variable frequency drive or EC motor for variable speed operation.
- 14. Power circuit for chemical treatment
- 15. Make-up water system and metering
- 16. Freeze protection measures
- 17. Drainage system for the basin

PART 1 GENERAL

1.1 REFERENCES

NOTE: This paragraph is used to list the publications cited in the text of the guide specification. The publications are referred to in the text by basic designation only and listed in this paragraph by organization, designation, date, and title.

Use the Reference Wizard's Check Reference feature when you add a Reference Identifier (RID) outside of the Section's Reference Article to automatically place the reference in the Reference Article. Also use the Reference Wizard's Check Reference feature to update the issue dates.

References not used in the text will automatically be deleted from this section of the project specification when you choose to reconcile references in the publish print process.

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ACOUSTICAL SOCIETY OF AMERICA (ASA)

ASA S1.13 (2020; R 2025) American National Standard
Measurement of Sound Pressure Levels in Air

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

ANSI/AHRI 495 (2005) Performance Rating of Refrigerant
Liquid Receivers

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ANSI/ASHRAE 15 & 34 (2022) ASHRAE Standard 15-Safety Standard
for Refrigeration Systems and ANSI/ASHRAE
Standard 34-Designation and Safety
Classification of Refrigerants

ASHRAE 64 (2020) Methods of Testing Remote
Mechanical-Draft Evaporative Refrigerant

Condensers

AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1 (2021) Safety in Welding, Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M (2022) Standard Specification for Gray Iron Castings

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A653/A653M (2025a) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM B117 (2025) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C67/C67M (2025) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile

ASTM D520 (2025) Standard Specification for Zinc Dust Pigment

ASTM D1784 (2025) Standard Classification System and Basis for Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds

ASTM D2996 (2023) Standard Specification for Filament-Wound "Fiberglass" (Glass-Fiber-Reinforced Thermosetting-Resin) Pipe

ASTM E84 (2025) Standard Test Method for Surface Burning Characteristics of Building Materials

COOLING TECHNOLOGY INSTITUTE (CTI)

CTI ATC-105 (2019) Acceptance Test Code for Water Cooling Towers

CTI STD-111 (2024) Gear Speed Reducers for Application on Industrial Water Cooling Towers

CTI Std-137 (2017) Fiberglass Pultruded Structural Products for Use in Cooling Towers

CTI Std-201 (2011) Standard for the Certification of Water Cooling Tower Thermal Performance

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 00001 (2024) Motors and Generators

NEMA MG 11 (1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2026; TIA 26-1; ERTA 26-1; TIA 26-2; TIA 26-3; TIA 26-4; TIA 26-5; TIA 26-6; TIA 26-7; ERTA 26-2; ERTA 26-3) National Electrical Code

NFPA 214 (2026) Standard on Water-Cooling Towers

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE J534 (2021) Lubrication Fittings

UL SOLUTIONS (UL)

UL 723 (2018; Reprint Jun 2025) UL Standard for Safety Test for Surface Burning Characteristics of Building Materials

1.2 SUBMITTALS

NOTE: Review Submittal Description (SD) definitions in Section 01 33 00 SUBMITTAL PROCEDURES and edit the following list, and corresponding submittal items in the text, to reflect only the submittals required for the project. The Guide Specification technical editors have classified those items that require Government approval, due to their complexity or criticality, with a "G." Generally, other submittal items can be reviewed by the Contractor's Quality Control System. Only add a "G" to an item if the submittal is sufficiently important or complex in context of the project.

For Army projects, fill in the empty brackets following the "G" classification, with a code of up to three characters to indicate the approving authority. Codes for Army projects using the Resident Management System (RMS) are: "AE" for Architect-Engineer; "DO" for District Office (Engineering Division or other organization in the District Office); "AO" for Area Office; "RO" for Resident Office; and "PO" for Project Office. Codes following the "G" typically are not used for Navy and Air Force projects.

The "S" classification indicates submittals required

**as proof of compliance for sustainability Guiding
Principles Validation or Third Party Certification
and as described in Section 01 33 00 SUBMITTAL
PROCEDURES.**

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Open Cell Cooling Towers; G, [_____]

Posted Instructions; G, [_____]

Demonstrations; G, [_____]

Verification of Dimensions; G, [_____]

Remote Evaporatively-Cooled Condensers; G, [_____]

Closed Circuit Coolers; G, [_____]

SD-06 Test Reports

- [Packaged Cooling Tower - Installation Instructions; G, [_____]
-][Field-Erected Cooling Tower - Installation Instructions; G, [_____]
-][Packaged Cooling Tower - Field Acceptance Test Plan; G, [_____]
-][Field-Erected Cooling Tower - Field Acceptance Test Plan; G, [_____]
-][Packaged Cooling Tower - Field Acceptance Test Report; G, [_____]
-][Field-Erected Cooling Tower - Field Acceptance Test Report; G, [_____]
-][Start-Up Testing Report; G, [_____]

] SD-07 Certificates

Service Organization

Cooling Tower

SD-08 Manufacturer's Instructions

- [Packaged Cooling Tower - Installation Instructions
-][Field-Erected Cooling Tower - Installation Instructions

] SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals

Remote Evaporatively-Cooled Condensers

1.3 SAFETY REQUIREMENTS

NOTE: Catwalk, ladder and guardrail may be required. If so, select the applicable item and delete the others and indicate on drawings the selected item. If not applicable, delete the entire sentence within the brackets.

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.[[Catwalk,][ladder,][and guardrail]must be provided where indicated and in accordance with[Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS][Section 05 51 33 METAL LADDERS][Section 05 52 00 METAL RAILINGS][Section 05 51 00 METAL STAIRS].]

1.4 DELIVERY, STORAGE, AND HANDLING

Stored items must be protected from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation must be the Contractor's responsibility. Any materials found to be damaged must be replaced at the Contractor's expense. During installation, piping and similar openings must be capped to keep out dirt and other foreign matter.

1.5 PROJECT/SITE CONDITIONS

1.5.1 Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. The Contractor must carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and must arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

PART 2 PRODUCTS

2.1 STANDARD COMMERCIAL PRODUCTS

Materials and equipment must be standard commercial catalogued products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. The standard products must have been in satisfactory commercial or industrial use in field service for two years prior to bid opening. The two year use must include applications of equipment and materials under similar circumstances and of similar size. Products having less than a two year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of

the manufacturer's factory tests, can be shown. This 6000 hour record must not include any manufacturer's prototype or factory testing. Records of satisfactory field use must be completed by a product that had been, and presently is, sold, or offered for sale on a commercial market through the following copyrighted means: advertisements, manufacturer's catalogs, or brochures. Products must be supported by a [service organization](#). System components must be environmentally suitable for the indicated locations.

2.2 MANUFACTURER'S STANDARD NAMEPLATES

Major equipment including cooling towers, cooling tower gear drive assemblies, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life. Plates must be fixed in prominent locations.

2.3 ELECTRICAL WORK

NOTE: Where motor starters for mechanical equipment are provided in motor-control centers, the references to motor starters must be deleted.

Show the electrical characteristics, motor starter type(s), enclosure type, and maximum rpm on the drawings in the equipment schedules.

Where reduced-voltage motor starters are recommended by the manufacturer or required otherwise, specify and coordinate the type(s) required in Section **26 20 00 INTERIOR DISTRIBUTION SYSTEM**. Reduced voltage starting is required when full voltage starting will interfere with other electrical equipment and circuits and when recommended by the manufacturer. Where adjustable speed drives (ASD) are specified, reference Section **26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS**. The methods for calculating the economy of using an adjustable speed drive is described in UFC 3-520-01, "Interior Electrical Systems".

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section **26 20 00 INTERIOR DISTRIBUTION SYSTEM**. Manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown, must be provided. For packaged equipment, the manufacturer must provide controllers including the required monitors and timed restart.
- b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with **NEMA MG 11**.
- c. For polyphase motors, provide squirrel-cage medium induction motors,

including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with NEMA MG 00001.

NOTE: Bracketed sentence "Motor bearings..." to be used for Army projects only.

- d. Provide motors in accordance with NEMA MG 00001 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Motors must be rated for continuous duty with the enclosure specified. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. [Motor bearings must be fitted with grease supply fittings and grease relief to outside of the enclosure.] Motor enclosure type may be either TEAO or TEFC.
- e. [Where two-speed motors are indicated, variable-speed controllers may be provided to accomplish the same function.][Use adjustable frequency drives for all variable-speed motor applications.]Provide variable frequency drives for motors as specified in Section 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS.
- f. Provide inverter duty premium efficiency motors for use with variable frequency drives.

2.4 COOLING TOWER MATERIALS

2.4.1 Fiberglass Reinforced Plastic (FRP)

FRP components must be inert, corrosion resistant, and fire-retardant with a thickness of 3.66 kg per square meter 12 ounces per square foot. FRP components must contain an ultraviolet (UV) ray inhibitor as per CTI Std-137, Grade 1 or 3. Components manufactured of polystyrene are not permitted.

2.4.2 Zinc-Coated Steel

Components fabricated of zinc-coated steel must be not lighter than 16 gauge 1.613 mm 0.0635 inch steel, protected against corrosion by a zinc coating. The zinc coating must conform to ASTM A653/A653M, as applicable and have an extra heavy coating of not less than 760g per square meter 2.35 ounces per square foot of surface. Galvanized surfaces damaged due to welding must be coated with zinc rich coating conforming to ASTM D520, Type 1.

2.4.3 Polyvinyl Chloride (PVC) Formed Sheets

ASTM D1784, Type I, Grade 1 with a flame spread rating in accordance with ASTM E84, Class A.

2.4.4 High Density Polyethylene (HDPE)

Components manufactured from HDPE must be seamless with a minimum

thickness of 10 mm 0.375 inch. The material must have the appropriate inhibitors to protect the component from any UV degradation. Tanks and cooling tower shells must be seamlessly molded to minimize water loss/consumption.

2.4.5 Stainless Steel Sheets

NOTE: Designer should consider location (non-coastal, coastal, extremely corrosive areas) when specifying corrosion protection. Additional lead time and cost may be a factor. Designer must consult local review team for guidance on corrosion protection. Type 316 stainless steel is generally needed in coastal areas, while stainless steel type 304 is generally acceptable in all other areas. For other corrosive environments review with equipment vendors.

Components manufactured from stainless steel must be stainless steel type[304][316]. Provide welded stainless steel seams at panels, basins, and other components.

2.4.6 Concrete

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE. Exposed concrete must be rub-finished for smooth and uniform surfaces free of form marks and defects. Honeycomb concrete is not permitted.

2.4.7 Hardware

Bolts must be cadmium-plated, zinc-coated steel, or Type 304 stainless steel. Each bolt must be provided with neoprene and cadmium-plated steel washers under the heads. Nails must be silicon bronze, commercial bronze, or stainless steel. Hardware must meet the salt-spray fog test as defined by ASTM B117. Angle brackets and similar parts must be zinc-coated steel. Zinc coatings must conform to ASTM A153/A153M and[ASTM A123/A123M], as applicable, and must have an extra heavy coating of not less than 760g per square meter 2.35 ounces per square foot of surface. No signs of corrosion must be evident after 1,000 hours continuous exposure to a 5 percent salt spray.

2.5 OPEN CELL COOLING TOWERS

NOTE: Edit and update the specification, including adding features as necessary, per actual equipment selection and project requirements.

2.5.1 Factory Assembled Towers

2.5.1.1 Description

The cooling tower must be of the[induced mechanical draft][or forced mechanical draft] type. The cooling tower must include frames and casings, louvers, drift eliminators, partitions, windbreak baffles, drift-check walls, cold water basin equipment, fans and fan walls,

blowers, drives, electric motors, access doors,[working platforms,]
inspection plates, and panels.

2.5.1.2 Construction

Tower must be constructed to withstand a wind pressure of not less than **1.44 kilopascal (kPa)** **30 psf** on any external surface. Fan deck must be constructed to withstand a live load of not less than **2.87 kPa** **60 psf** in addition to the concentrated or distributed loads of equipment mounted on the fan deck.[A 15 percent increased loading must be included for ice or snow load.]

The hot water distribution system must be of the open basin gravity feed type or the pressurized spray header type design.

2.5.1.3 Tower Frame and Louvers

Provide frame constructed from [galvanized steel][_____]. Intermediate structural members must be provided for rigidity and support of casings, louvers, fill, distribution systems, fan decks, and other equipment. Inlet air louvers must permit free air passage but no splash out, and must be designed to prevent debris and sunlight from entering the cold water basin.

[2.5.1.4 Air Inlet and Discharge Connections

On forced draft centrifugal type units, the air inlet and discharge connections must have flanged or lipped projections for connecting to ductwork.

]2.5.1.5 Fill

The fill must support expected loads without sag or failure and arranged to effectively break up the water. The fill must be manufactured and performance tested by the cooling tower manufacturer. The fill must be of the materials as specified. Polyvinyl chloride (PVC) fill is suitable for inlet temperatures to **51.7 degrees C** **125 degrees F** on cross flow type units and temperatures to **54.4 degrees C** **130 degrees F** on counterflow type units. Chlorinated polyvinyl chloride (CPVC) fill must be used for applications where inlet temperatures are greater than **51.7 degrees C** **125 degrees F**. Fill must be in accordance with **ASTM E84**, Class A.

2.5.1.6 Drift Eliminators

Provide drift eliminator sections designed and arranged to effectively trap water droplets entrained in the discharge airstream. Sections must be assembled in easily removable sections for[forced mechanical drift tower][and][counterflow induced mechanical draft tower]. Drift eliminators must be constructed of Polyvinyl chloride (PVC) in accordance with **ASTM E84**, Class A.

2.5.1.7 Cold Water Basin Equipment.

NOTE: Choose galvanized steel, except in corrosive environments, choose Type 304 stainless steel.

Include[galvanized steel][Type 304 stainless steel] sump with stainless

steel removable screen and vortex breaker, float valves, and necessary pipe connections and fittings within the tower.[Provide float valves with adjustable arms. Valve sizes larger than 13 mm 1/2 inch pipe size must be the balanced piston type. Valve seats and disks must be replaceable.][Electronic water level control must be provided.]

Provide cold water basins and casings suitably sealed and flashed at joints and connections to ensure watertight construction.

2.5.1.7.1 Electric Basin Heater

NOTE: Coordinate ambient temperature with Electric Basin heater paragraph below.

Heater must be the electric immersion type with water-tight junction boxes mounted in the basin with sufficient capacity to maintain the basin water temperature above 4.4 degrees C 40 degrees F at an ambient temperature of [_____] degrees C degrees F. Heater must be complete with control thermostat, transformer, contactor, and low water level heater protection.

2.5.1.8 Fans, Blowers, and Drives.

NOTE: Designer should consider reduced maintenance requirements associated with direct drive assemblies.

The towers must have axial propeller-type fans having not less than four aluminum alloy or glass-reinforced polypropylene blades or squirrel-cage, centrifugal-type blowers, as applicable. Fans and blowers must be designed and constructed to withstand 50 percent overspeed above normal maximum operating speeds.

If belt drives are utilized, multi-grooved solid back single belt design must be used to avoid uneven belt stretch. Adjustment must be provided for belt tension and drive centers. Belt drives must be designed and constructed for 150 percent overload. Sheaves located in the airstream must be corrosion-resistant material. Shafting for gear drives must have flexible-type couplings requiring no lubrication. The gear assemblies must be enclosed in an oil-filled housing provided with fill and drain plugs.

2.5.1.9 Tower Piping

Piping must be schedule 40 PVC and conform to ASTM D2996. Piping supports must be corrosion resistant hangers and supports that are designed to resist movement during operation and shipment. Fittings for other piping materials must be of the same material or equal and of the same class and grade as the pipe.

2.5.1.10 Electric Motors

Requirements are specified in paragraph ELECTRICAL WORK.

[2.5.1.11 Vibration Cutout Switch.

Provide[mechanical vibration cutout switch][electronic vibration cutout

switch with auxiliary contacts] in a protected position and most effective location, interlocked with the fan wiring to electrically open the motor circuit under excessive fan vibration.

]2.5.1.12 Performance

The factory assembled tower must have Cooling Tower Institute certification that, in accordance with CTI Std-201, the cooling tower must perform thermally at the rating published by the tower manufacturer in his copyrighted literature.

2.5.1.13 Sound Power Level

NOTE: The numbers shown in Tables I & II are ranges of acceptable/recommended sound power levels.

Sound power levels, in decibels (dB) with a reference pressure of 0.0002 microbars, of the cooling tower must be not greater than the maximum permitted dB levels for the designated octave band as set forth in Table I or Table II. The sound power level data for the cooling tower must have been verified in tests conducted in accordance with ASA S1.13.

Table I. Sound Power Level for Induced Mechanical Draft Type								
Octave Band (Hz)	63	125	250	500	1000	2000	4000	8000
Sound Power Level (dB)	112	112	110	108	102	98	93	90

Table II. Sound Power Level for Forced Mechanical Draft Type								
Octave Band (Hz)	63	125	250	500	1000	2000	4000	8000
Sound Power Level (dB)	112	112	110	108	102	98	93	90

2.5.1.14 Drift Loss

Drift loss must be not greater than 0.005 percent of the water circulated.

2.5.2 Lubrication

The lubricating points must be extended to the outside of the unit for easy accessibility. Hydraulic lubrication fittings must be in accordance with SAE J534. Where use of high pressure lubricating equipment, 6894 kPa 1000 psi or higher, will damage grease seals or other parts, a suitable warning must be affixed to the equipment in a conspicuous location.

2.5.3 Factory Finish System

NOTE: Galvanized metal is the standard finish for most manufacturers. Provide finish systems with 3,000 hour salt spray compliance for seacoast installations

[Factory painting system][Galvanized metal]must have been proven to withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Equipment located in a seacoast environment must withstand 3,000 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with **ASTM B117**. For salt-spray fog test, the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond **3 mm 0.125 inch** on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above **50 degrees C 120 degrees F**, the factory painting system must be designed for the temperature service and must have been proven to pass the specified salt-spray test.

[2.5.4 Field-Assembled Cooling Towers

Factory fabricated, factory-assembled towers which are shipped to the job site in separate cells or modules must be provided with all appropriate manufacturer's hardware for assembly in the field. Factory fabricated, field-assembled towers must be assembled and adjusted at the job site by a factory representative.

2.5.4.1 Framework, Casing, and Supports

NOTE: Packaged type cooling towers are typically constructed to withstand a 1.4 kPa (30 psf) wind load.

Towers must be designed and constructed to withstand a wind pressure of not less than **[1.4][] kPa [30][] pound-force per square foot (psf)** on external surfaces.[A 15 percent increased loading must be included for ice or snow load.][Air inlet and discharge terminations must have flanged or lipped projections for connecting ductwork.] Framework, structural supports, and equipment supports must be[zinc-coated steel,][Type 304 stainless steel,][air-entrained concrete][or][FRP]. Casing (exterior enclosing walls) must be constructed of[zinc-coated steel][Type 304 stainless steel][air-entrained concrete][or][FRP]. Materials provided for framework, casings and equipment supports must be compatible. Structural supports must be provided in accordance with the recommendations of the manufacturer of the tower unless otherwise indicated.[Cold-pour concrete joints in vertical walls must have a continuous water-stop stripping of molded polyvinyl plastic (**150 mm 6 inch** dumbbell).]

2.5.4.2 Foundations

NOTE: For the design of a tower foundation, indicate the location, the size, the reinforcement requirements, etc. necessary for a cooling tower available from three commonly known manufacturers. For small retrofit type jobs the designer may choose to show the general layout of the foundation and rely on the Contractor to design and construct the foundation based on the cooling tower to be provided. Delete the last two sentences of the paragraph if the foundation is not to be designed by the Contractor.

Cooling tower foundations must meet the requirements of the cooling tower manufacturer and wind and seismic loads, wind and seismic loads and be as indicated. Foundation design must be based on the load conditions and soil bearing value indicated. Foundation calculations must be submitted with the equipment drawings.

2.5.4.3 Stairways and Ladders

Provide stairs, 60-degree ship ladders or straight-rung ladders of standard design, starting at[ground][roof] level and extending as high as required to gain access to fan decks and water distribution systems. Stairways and ladders must be hot-dip, zinc-coated steel. Ladders higher than 3.66 meters 12 feet must have a safety cage.

2.5.4.4 Hand Railings

Steel hand railings must be not less than 1067 mm 42 inches high around the exterior of each working surface that is 3.66 m 12 feet or more above the ground, roof, or other supporting construction. Railings must be not smaller than 32 mm 1-1/4 inch zinc-coated steel pipe with standard zinc-coated steel railing.

2.5.4.5 Access Doors

Each tower must be provided with access doors at grade level to provide entry to the interior for service maintenance without removal of the fill. Doors must be provided on each end wall of each cooling tower cell. Frame and brace access doors to prevent damage when opening and closing. Doors must be located adjacent to float controls.

2.5.4.6 Louvers

Air inlets for each cooling tower must be provided with individually removable louvers arranged to prevent the escape of water. Louvers must be constructed of[PVC][fiberglass reinforced polyester][zinc-coated steel][Type 304 stainless steel][FRP]. Materials provided for casings and louvers must be compatible; one material must not produce stains upon the other. Air intakes must be provided with 25 mm 1 inch zinc-coated steel mesh.

2.5.4.7 Fan Deck and Cylinder

Each fan must be mounted in a fan cylinder (or stack) to elevate the fan discharge air. Total extension height must not exceed the fan diameter. Each fan cylinder must be provided with a zinc-coated steel, 12 gauge 2.753 mm 0.108 inch wire mesh securely mounted to the top of the cylinder

in accordance with manufacturer's recommendations. Fan decks must be designed to withstand a live load of not less than [1.9][2.9] kPa [40][60] psf in addition to the concentrated or distributed loads of equipment mounted on the fan decks.[Fan deck and cylinders must be constructed of zinc-coated steel, Type 304 stainless steel, or FRP and be compatible with the entire tower construction.][Fan deck must be constructed of precast, reinforced lightweight concrete, in multiple sections, forming a complete, vibration-free base for mounting fan, speed reducer, drive shaft, motor, and fan stacks. Fan cylinders (or stacks) must be constructed of precast, reinforced lightweight concrete in multiple sections, constrained with bands of zinc-coated steel conforming to ASTM A123/A123M, not less than 3 by 75 mm 1/8 by 3 inches, and bolted to form a compressive load on stack perimeter. Fan cylinder must be secured in place on the fan deck with Class A mortar.]

2.5.4.8 Fans

NOTE: When the density of the ambient air to be handled by the fans differs substantially from the density of the standard air value of 1.2 kg per cubic m 0.075 pound per cubic foot at 21 degrees C 70 degrees F and 101 kPa 29.92 inches mercury, the density of the air and the elevation above mean sea level must be shown on the drawings.

Fans must be the[centrifugal][or][adjustable-pitch propeller] type, constructed of zinc-coated steel, Type 304 stainless steel, aluminum or an aluminum alloy, or FRP. Propeller type fans must have a maximum tip speed of 330 m/minute 10,800 fpm. Fan blade assembly must be both statically and dynamically balanced after assembly of the cooling tower. Fan hub must be constructed of[zinc-coated steel][stainless steel][cast aluminum] with adequate surface protection against corrosion. Complete fan assembly (fan and mounting) must be designed to give maximum fan efficiency and long life when handling saturated air at high velocities. Each cooling tower fan must be provided with a ball and pedestal type vibration limit switch which must stop the corresponding fan motor in the event of sensing excessive fan vibration.

2.5.4.9 Speed Reducers Gears and Drive Shaft

NOTE: Double reduction gear reducer should be considered where low noise requirement is a factor.

Speed reducer gears must be rated in accordance with CTI STD-111. Gear reducers must be of the[spiral bevel, single reduction][spiral or helical, double reduction] type. Reducer must be mounted in accordance with manufacturer's recommendations. Each reducer must be provided with an oil level cutoff switch interlocked to the fan motor. Each reducer must be provided with an oil level sight glass, fill, drain, and vent lines located in a readily accessible position. Drive shafts must be the full floating type with flexible couplings at both ends and have a service factor of 1.0 or greater. Drive shafts must be of stainless steel, fitted each end with flexible couplings (stainless steel plate type). Each drive shaft must be provided with a galvanized steel guard, to prevent damage to surrounding equipment in case of shaft failure. Provision must be made

for lubrication of all bearings. Bearings must be accessible to the extent that each bearing can be lubricated without dismantling fan.

2.5.4.10 Electric Motors

NOTE: Delete the last sentence if inapplicable.

Consider the following for energy efficiency in cooling towers:

Induced draft fans, VFD's and designing to 0.4 percent wet bulb temperature.

Each motor must be a[single speed][two speed][variable speed], totally enclosed, insulation Class B, NEMA Design B, continuous-rated type which conforms to **NEMA MG 00001**. Motors must have[open][drip proof][totally enclosed][explosion proof] enclosures and be located outside the discharge airstream. Motors must be mounted according to manufacturer's recommendations.[Two-speed motors must have a single winding with variable torque characteristics.][Motors for variable speed application must be inverter-duty type.] Motors must be provided specifically for either pump or fan application and must comply with the requirements of paragraph ELECTRICAL WORK.

2.5.4.11 Cold Water Basin

NOTE: Choose zinc-coated steel, except in corrosive environments, choose Type 304 stainless steel or FRP.

Basin must be completely watertight and constructed of[zinc-coated steel][Type 304 stainless steel][FRP]. Basin must be constructed and installed to ensure that air will not be entrained in outlets when operating and no water will overflow on shutdown. Each individual sump must be provided with an individual outlet. Each outlet must be provided with a **13 mm 1/2 inch** stainless steel wire mesh, securely mounted to prevent trash from entering the outlet. Each basin must be provided with overflow and drain valve connections. Each basin must be provided with a float-controlled, makeup water valve as indicated. The makeup water must discharge not less than **50 mm 2 inches** or two pipe diameters, whichever is greater, above the top of the basin.

2.5.4.12 Electric Basin Heater

Heater must be the electric immersion type with water-tight junction boxes mounted in the basin with sufficient capacity to maintain the basin water temperature above **4.4 degrees C 40 degrees F** at an ambient temperature of [____] **degrees C degrees F**. Heater must be complete with control thermostat, transformer, contactor, and low water level heater protection.

2.5.4.13 Hot Water Distribution System

NOTE: The gravity-flow type distribution system is the system of choice. Pressurized-flow type systems will typically only be specified for field-erected,

counterflow type towers

Piping connecting to a cooling tower must be externally supported, independent of the tower structure and piping.

Water distribution must be the[gravity-flow][pressurized-flow] type system which distributes water evenly over the entire fill surface. Each tower cell must be designed so that a water flow of 140 percent capacity will not cause overflowing or splashing. The distribution system for each cell must include adjustable flow control valves. The entire distribution system must be self-draining and non-clogging. Piping must be either cast iron, ductile iron, threaded-glass-fiber reinforced epoxy pipe, polypropylene, PVC or Schedule 80 black steel.

- a. Gravity-Flow System: System must be provided with open basins which include a splash box or baffles to minimize splashing of incoming hot water and holes that evenly distribute the water over the entire decking area. Holes used in a water basin must be provided with ceramic or plastic orifice inserts.
- b. Pressurized-Flow System: System must include piping, fittings, branches, and spray nozzles. Spray nozzles must be schedule 40 PVC. Nozzles must be cleanable, non-clogging, removable, and spaced for even distribution.
- c. Basin Cover: Hot water distribution basins must be provided with the tower manufacturer's standard removable,[zinc-coated galvanized steel][304 stainless steel][FRP] covers. Covers must prevent airborne debris from entering the basin.

]2.5.5 Drift Eliminators

NOTE: Per ICC IgCC, 601.3.2.3(c) 6.3.2.3(c), as invoked from UFC 1-200-02, cooling towers must be equipped with efficient drift eliminators that achieve drift reduction to a maximum of 0.002 percent if the recirculated water volume for counterflow towers and 0.005 percent of the recirculated water flow for crossflow towers.

Delete the last set of bracketed sentences if a field-erected type tower is not specified.

Eliminators must be provided in the tower outlet to limit drift loss to not over[0.002][0.005] percent of the circulating water rate. Eliminators must be constructed of polyvinyl chloride (PVC).[Eliminators sections must be supported on PVC or FRP tee sections. Tee sections must be suspended with 6.35 mm 1/4 inch brass rods connected to stainless steel clips embedded in the bottom side of the roof deck at the time of casting. Stainless steel clips must be supplied by cooling tower manufacturer for installation by Contractor at time of roof deck pour. Eliminators may be supported by brass or stainless steel suspension rods from the fan deck or supported directly on concrete beams.]

2.5.6 Cold Water Basin Equipment.

Include sump with removable screen and vortex breaker, float valves, and necessary pipe connections and fittings within the tower. Provide float valves with adjustable arms. Valve sizes larger than 13 mm 1/2 inch pipe size must be the balanced piston type. Valve seats and disks must be replaceable.[Electric water level control must be provided.]

Provide cold water basins and casings suitably sealed and flashed at joints and connections to ensure watertight construction.

2.5.7 Fill (Heat Transfer Surface)

NOTE: Typically, both the splash or film type tower fill will be allowed. Film type fill must not be allowed where there is a highly likely possibility that the circulating water will become contaminated with debris (leaves, etc.). Debris in the circulating water will significantly impact the efficiency of a tower with film type fill because of the close spacing of the film material. Note that hot water distribution basin covers will typically prevent most debris from ever getting to the fill material.

The most predominant fill material is PVC formed sheets. PVC formed sheets, zinc-coated steel will be the typical choices for fill material. Aluminum or stainless steel fill must only be specified where either high inlet water temperatures or fireproof construction are concerns. PVC formed sheets must not be provided when the inlet water temperature exceeds 125 degrees F. Tile fill will only be considered on field-erected type towers where economically justified.

Delete the bracketed sentences at the end of the paragraph if tile type fill material is not specified.

Tower fill must be the[splash][or][film] type. Fill material must be free to expand or contract without warping or cracking. No plasticized wood cellulose must be provided for fill material. Fill must be removable or otherwise made accessible for cleaning. Space supports must be corrosion resistant and must prevent warping, sagging, misalignment, or vibration of the fill material. Fill material and supports must be designed to provide for an even mixing of air and water. Fill material must be constructed of[aluminum][stainless steel][tile of multi-cell design, set without mortar][PVC formed sheets, zinc-coated steel] in a pattern, and of sufficient height to meet the performance specifications.[Tile fill must be vitreous, with a low water absorption that must pass a freeze-thaw test conducted in accordance with ASTM C67/C67M. Tile fill must have a minimum crushing strength of 13.8 MPa 2,000 psi over the gross area of the tile when the load is applied parallel to the cells as tested in accordance with ASTM C67/C67M. Cast iron tee section lintels supporting the tile fill must conform to ASTM A48/A48M, Class 25, 3.2 mm 1/8 inch additional thickness for

corrosion. Lintels must be designed with a safety factor of 2 minimum.]

2.5.8 Fire Safety

NOTE: Locate the tower in accordance with NFPA 214, and determine the extent and type of fire protection required for all size towers using the factors indicated in NFPA 214. Indicate fire protection system requirements in the design documents. Coordinate with fire protection specs and make sure any additional requirements are added to Section 21 13 16 DRY PIPE SPRINKLER SYSTEMS, FIRE PROTECTION as needed for cooling towers.

Towers must conform to NFPA 214. Fire hazard rating for plastic impregnated materials must not exceed 25. Plastics must not drip or run during combustion. Fire hazard ratings must be in accordance with ASTM E84, Class A and UL 723.[Refer to Section 21 13 16 DRY PIPE SPRINKLER SYSTEMS, FIRE PROTECTION for fire suppression systems requirement associated with the equipment.]

2.5.9 Meters and Controls

Tower must be provided with makeup and blowdown meters, conductivity controller[, electric immersion heaters and controls], and water level control system with overflow alarm. Include a controller with a NEMA 4 control panel containing[unfused disconnect switch][, interlocks and relays][, pilot lights and push buttons][, and must integrate into the building automation system specified in[Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC[Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS,][Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION][or][Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS][, and must integrate into the existing building automation system]].

2.6 CLOSED CIRCUIT COOLERS

NOTE: Closed circuit cooling towers are generally used for heat pump, other water cooled systems with smaller heat exchangers in the equipment, or smaller tonnage systems.

Edit and update the specification including adding features as necessary, per actual equipment selection and project requirements.

2.6.1 Description

Provide a closed circuit cooler factory assembled and tested induced draft unit with vertical air discharge, complete with vertical discharge axial fan, motors, and drive system, heat exchanger, fill and drift eliminators, louvers, heat-loss closure dampers, spray water system, and accessories and rigging supports. The tower and piping systems are to be arranged for retention and storage of spray water in the tower basin.

2.6.2 Casing and Finish

Construct the fan section (up to top of intake louvers) of heavy gage stainless steel designed for a minimum uniform live load of 292 kgf/m 260 psf and construct casing of hot-dip galvanized steel. Apply corrosion protection system to the outside of galvanized surfaces. Construct non-stainless metal components of mill hot-dip galvanized steel. Coat component edges and welds with a 95 percent pure zinc-rich compound. Preparation for coating must include degreasing, cleaning, and light surface burnishing. The coating must be suitable for field repair with the same original coating material applied in the same manner.

2.6.3 Louvers

Provide air inlet louver that is UV resistant PVC. Mount louvers in removable frames for maintenance access to the cold water basin. Provide louvers with a minimum of two changes in air direction to prevent splash out and block direct sunlight.

[2.6.4 Discharge Hood with Positive Closure Dampers

Provide unit with discharge hood, positive closure dampers, and 120-volt spring-return normally closed actuator for reduction of heat loss during idle periods of winter time operation. Construct the discharge hood and dampers of [hot dipped galvanized steel] [stainless steel]. Equip hoods with access panels to facilitate maintenance on the eliminators and water distribution system. Factory assemble the dampers, damper actuator, and linkage.

]2.6.5 Heat Exchanger

The coil must be [steel, encased in steel framework with the entire assembly hot-dip galvanized after fabrication] [copper] [stainless steel]. Arrange tubes in a self-spacing, staggered pattern in the direction of airflow for maximum heat transfer efficiency and minimum pressure drop, without the use of additional spacers between the coil tubes. Design coil with sloping tubes for free drainage of liquid and test to 2413 kPa 350 psi air pressure under water.

2.6.6 Drift Eliminators

Construct eliminators of inert PVC in easily handled sections. Provide an eliminator design that incorporates three changes in air direction to ensure complete removal of entrained moisture from the discharge airstream.

2.6.7 Cold Water Basin and Intermediate Basins

The cold water basin, intermediate basins (when applicable), and any steel components in contact with the water must be constructed of Type 304 stainless steel. Slope the basin area under the fill to facilitate cleaning. Factory weld all basin seams for tight construction. Standard basin accessories must include louver access, overflow, drain, Type 304 stainless steel strainers, removable anti-vortexing device made of 304 stainless steel to prevent air entrainment, and corrosion resistant make-up valve with float that allows for easy adjustment of the operating water level.

2.6.7.1 Electric Basin Heater

Electric pan heater package consists of electric immersion heaters, heater thermostat, and low water cutout, all installed in pan. Size heaters to maintain plus 5 degrees C 40 degrees F basin water at [_____] degrees C degrees F ambient air temperature with the fans off at design conditions indicated on drawings. Control the heaters with a thermostat and provide water cutout to prevent heaters from cycling on unless they are completely submerged. Provide heater contactor and wiring under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.6.8 Fans and Drives

Provide statically balanced heavy duty axial propeller type fans. Construct fans with aluminum alloy blades and install in a closed fitted cowl with venturi air inlet. Fan drive must be multi-groove, solid V-belt type with taper lock sheaves designed for 150 percent of nameplate kW HP. Fan and motor sheaves must be aluminum alloy construction. Belt adjustment must be accomplished from exterior of unit.

2.6.9 Piping

The system must provide a water flow rate of not less than .3846 l/sec 6 gpm over each square foot of unit face area to ensure proper flooding of the coil. Construct spray header of Schedule 40 polyvinyl chloride (PVC) pipe for corrosion resistance. Provide removable spray branches for cleaning. Distribute water over the entire coil surface by spray nozzles (381 by 8 mm 15 by 5/16 inch orifice) with internal sludge ring to eliminate clogging. Thread nozzles into spray header to provide easy removal for maintenance.

2.6.10 Electric Motors

Fan Motors: Motors must be totally enclosed, ball bearing type, and suitable for outdoor service. Motors 1 Hp and greater must be the premium efficiency type in accordance with NEMA MG 00001. Provide fan motor speed control for motors 7.5 hp or larger.

2.6.11 Water Recirculation Pump

Provide a close-coupled, centrifugal type pump with mechanical seal, installed vertically at the factory to allow free drainage at shutdown.

2.6.12 Drift Loss

Design the tower such that maximum drift rate is less than 0.005 percent of the circulating water rate.

2.6.13 Sound Power Level

NOTE: The numbers shown in Tables I & II are ranges of acceptable/recommended sound power levels.

Sound power levels, in decibels (dB) with a reference pressure of 0.0002 microbars, of the closed circuit cooler must be not greater than the maximum permitted dB levels for the designated octave band as set forth in Table I or Table II. The sound power level data for the closed circuit

cooler must have been verified in tests conducted in accordance with
ASA S1.13.

Table I. Sound Power Level for Induced Mechanical Draft Type								
Octave Band (Hz)	63	125	250	500	1000	2000	4000	8000
Sound Power Level (dB)	112	112	110	108	102	98	93	90

Table II. Sound Power Level for Forced Mechanical Draft Type								
Octave Band (Hz)	63	125	250	500	1000	2000	4000	8000
Sound Power Level (dB)	112	112	110	108	102	98	93	90

2.6.14 Vibration Cutout Switch

Provide[mechanical vibration cutout switch][electronic vibration cutout switch with auxiliary contacts] in a protected position and most effective location, interlocked with the fan wiring to electrically open the motor circuit under excessive fan vibration.

2.7 REMOTE EVAPORATIVELY-COOLED CONDENSERS

Condenser must be rated and tested in accordance with the requirements of ASHRAE 64. Condenser must include fans, water pump with suction strainer, electric motor and drive equipment, water eliminators if required, condensing coil, liquid receiver if required, water pan or sump, spray nozzles or water-distribution pan, water strainer, water make-up assembly, bleeder with flow valve of the needle valve type sized for the flow required or a fixed orifice, enclosure with suitable access doors, and air-inlet and outlet openings. No water may carry over into the unit discharge outlet.

2.7.1 Condenser Casing

NOTE: Standard casing construction is galvanized steel. For excessively corrosive atmospheres, Type 304 stainless steel should be considered.

Enclosure must be constructed of not lighter than 18 gauge 1.311 mm 0.516 inch[hot-dip galvanized steel][304 stainless steel], reinforced and braced. Access doors or panels suitably sized and located must be provided for access to water nozzles or distribution pan, coils, and valves for cleaning, repair, or removal of the item. Access doors or panels must be gasketed with synthetic rubber, or equivalent gasket material, and locked in place with thumb screws or catches. One-half inch mesh hot-dip galvanized steel or copper air-inlet screens must be provided on each air inlet.

2.7.2 Refrigerant Section

Condenser coil must be constructed of unfinned copper or steel tubes hot-dip galvanized after fabrication. The receiver must be welded steel and must be fitted and tested in accordance with ANSI/AHRI 495. A refrigerant charging valve must be installed in the liquid line between the receiver cut-off valve and the expansion device. Refrigerant section must be tested in accordance with ANSI/ASHRAE 15 & 34 for the refrigerant employed in the system. CFC-based refrigerants are prohibited.

2.7.3 Fans

Fans must be centrifugal or propeller type as best suited for the application. Fans must be direct or V-belt driven. Belt drives must be completely enclosed within the unit casing or equipped with a guard. When belt drive is provided, an adjustable sheave to furnish not less than 20 percent fan-speed adjustment must be provided. The sheaves set must be matched and selected to provide the capacity indicated at the approximate midpoint of the adjustment. Fans must be statically and dynamically balanced. Fan motor must be totally enclosed type or open drip-proof and located within an enclosure to be fully protected from the weather.

2.7.4 Water Section

Water eliminators must be constructed of nonferrous metal, of an approved nonmetallic material, or of not lighter than 24 gauge 0.701 mm 0.0276 inch steel, hot-dip galvanized after fabrication. Spray nozzles must be brass non-clogging type designed to permit easy disassembly and must be arranged for easy access. Water pump must be bronze-fitted centrifugal or turbine type and may be mounted as an integral part of the evaporative condenser or remotely on a separate mounting pad. Pump suction must be fully submerged and provided with screened inlet. Water pan or sump must be constructed of not lighter than 14 gauge 1.994 mm 0.0785 inch steel, hot-dip galvanized after fabrication, or molded acid-resistant glass-fiber-reinforced polyester. Water distribution pan must be constructed of not lighter than 16 gauge 1.613 mm 0.0635 inch steel, hot-dip galvanized after fabrication. Joints must be watertight. Water pan or sump must be provided with drain, overflow, and make-up water connection with stop valve and float valve. A bleed line with a flow valve of the needle type sized for the flow required or fixed orifice must be provided in the pump discharge line and must be extended to the nearest drain for continuous discharge.

2.8 FABRICATION

NOTE: For equipment to be installed outdoors, adequate protection must be specified. Manufacturers must submit evidence that unit specimen have passed the specified salt spray fog test. A 500 hour test must be specified in a non-corrosive environment. A 3,000 hour test must be specified for seacoast environments.

Equipment and component items must have been proven to withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Equipment located in a seacoast environment must withstand 3,000 hours in a salt-spray fog test.

Salt-spray fog test must be in accordance with **ASTM B117**. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to **ASTM D520**, Type I.

2.9 SUPPLEMENTAL COMPONENTS/SERVICES

2.9.1 Condenser Water Piping and Accessories

Condenser water piping and accessories must be provided and installed in accordance with Section **23 64 26** CHILLED, CHILLED-HOT, AND CONDENSER WATER PIPING SYSTEMS.

2.9.2 Cooling Tower Water Treatment Systems

Cooling tower water treatment systems must be provided and installed in accordance with Sections **23 25 00** CHEMICAL TREATMENT OF WATER FOR MECHANICAL SYSTEMS and **23 64 26** CHILLED, CHILLED-HOT AND CONDENSER WATER PIPING SYSTEMS.

[2.9.3 Cooling Tower Passivation

NOTE: Cooling towers that have galvanized metal require special treatment during startup call passivation. Systems that are not properly treated may experience rapid corrosion of the new galvanized cooling towers, deposition of dirt and debris in the piping and heat exchanger equipment, microbiological control issues, and fouling from oil and grease.

Perform passivation of galvanized steel components in accordance with the cooling tower manufacturer's passivation guidelines.

]2.9.4 Cold Water Basin Sweeping System

Provide a side-stream basin sweeping and solids[separation][filtration] system, including packaged pumping and control system.

2.9.4.1 Basin Sweeping System

Fit the cold water basin with schedule 80 PVC sump sweeper piping complete with high-flow educator nozzles to facilitate in basin cleaning. Provide a system that contains two inlet connections and one outlet connection per basin, prepared for connection to a separately furnished separator.

[2.9.4.2 Separator

NOTE: Either a hydrocyclone separator or sand filter is needed to remove solids from the basin. The separator is most common, economical, and sufficient for most HVAC open tower condenser water applications. The sand filter provides a higher removal rate of smaller particles that may be more appropriate in dirtier environments.

Provide a packaged hydrocyclone separator and pump system that is factory

assembled with separator, high-head pump and motor(s), pre-strainer, pressure gauges, inlet and outlet manual isolation valves, skid, interconnecting face piping, control panel, and automatic purge. Incorporate a true tangential inlet and dumbbell body design in the hydrocyclone separator is required by the manufacturer through published results from an independent third-party testing laboratory. Standard test protocol of upstream injection, downstream capture, and separator purge recovery is allowed with 50-200 mesh particles to enable effective, repeatable results. Single pass test performance must not be less than 95 percent removal. In a single pass through the separator, given solids with a specific gravity of 2.6 and water at 1.0, performance, removal must be 98 percent of 74 microns and larger. Additionally, particles as fine as 5 microns with specific gravity greater than 2.6 must also be removed. Provide a hand-hole at the collection chamber access for either inspection or the removal of unusual debris on all separators with inlet/outlet connections 4 inches and larger. Locate the purge location at the lowest point of the separator accumulation chamber to prevent the buildup of unnecessary particulate within the separator. Provide automatic and manual air relief vents.

][2.9.4.3 Filtration

Provide a packaged sand filter and pump system that is factory assembled with filter media, system specific pump and motor(s), pre-strainer, inlet and outlet manual isolation valves, skid, interconnecting face piping, manual/auto air vent assembly, pressure gauges, control panel, and automatic backwash with electric valves and actuators and control panel. Provide a fully assembled sand media filter system to prevent the accumulation of troublesome solids in the open condenser water system. The influent fluid must be dispersed evenly over the filter media. Provide a system design that traps suspended solids in the pores formed between the media. The effluent fluid must pass from the vessel through the under-drain and return to the condenser water system. Provide a filter vessel, along with manway/hand-holes, fabricated of ASME stamped carbon steel with an epoxy lining and polyester coating. The vessel must be rated for 125 psi. Properly size the filter vessel inlet and outlet connections, drain connection, access ports, and vent fittings. Provide a filter unit with removable under-drain laterals, constructed of schedule 80 PVC. Provide a filter unit with a Schedule 80 PVC 3-point distribution over-drain for even flow across media, and a drain screen installed at the bottom of the filter vessel in order to retain media during scheduled draining of vessel. Provide a permanent sand type filter media with filter performance of 98 percent (nominal) of 10 micron particulate or larger and have achievable filtration down to 0.5 micron. Filter manufacturer must supply all required 10 micron media. Provide automatic and manual air relief vents. Provide backwash that are industrial grade 3-way ball type with a brass body and chrome plated brass ball for vessels 30 inches and smaller. For vessels 36 inches and larger, valves must be industrial grade cast iron lug style butterfly valves with stainless steel discs. Provide and utilize a motorized type electric actuator to cycle 3-way ball valves/butterfly valves between filter and backwash modes. Mechanically link the control valves.

][2.9.4.4 Pump Package

Provide cast iron bronze fitted, close coupled end-suction type pumps. Provide with 1/8-inch minimum perforated stainless steel basket pre-strainer with flange removable lid. Provide schedule 80 PVC packaged piping. Provide a structural steel framework constructed of carbon steel

that mounts the pump and[separator][filtration unit].

2.9.4.5 Power and Control

**NOTE: Separator units have a purge cycle while
filter units have a backwash cycle.**

UL listed and labeled NEMA type 4X polycarbonate enclosure with door disconnect switch for single point power connection, motor starter with short circuit/overload protection, step-down transformer, pressure differential switch, adjustable solid state timer and[purge timer][24-hour backwash clock], manual backwash push button, and remote start-stop with pump Hand-Off-Auto (HOA) switch. Control the pump by the Direct Digital Control (DDC) system. The pumps system is commanded off by the DDC system suspend the[purge][backwash] cycle.

]2.9.5 Temperature Controls

**NOTE: Modify this paragraph as required to
coordinate the central equipment controls with the
cooling tower system controls. In projects where
this section of the specifications is intended to
produce control equipment for existing systems, this
paragraph must be rewritten to properly integrate
the specified controls into the existing temperature
control system.**

**A sequence of control, a schematic of controls, and
a ladder diagram should be included on the drawings
for each cooling tower fan, chilled water pump,
condenser water pump, etc. in order to define the
overall system operation.**

Cooling towers must be fully coordinated with and integrated [into the temperature control system specified in[Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC][Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS][Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION][or][Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS]][into the existing air-conditioning system].

PART 3 EXECUTION

3.1 DEMONSTRATIONS

Contractor must conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total [_____] hours of normal working time and start after the system is functionally completed but prior to final acceptance tests. The training course must cover all of the items contained in the approved [Operation and Maintenance Manuals](#) as well as demonstrations of routine maintenance operations.

Provide a schedule, at least [2][_____] weeks prior to the date of the proposed training course, which identifies the date, time, and location

for the training.

3.2 INSTALLATION

Installation of cooling tower systems including materials, installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with NFPA 70, and in compliance with the manufacturer's written installation instructions, including the following:

- [(1) Packaged cooling tower - installation instructions
-][(2) Field-erected cooling tower - installation instructions

]3.2.1 Installation Instructions

Provide manufacturer's standard catalog data, at least [5][_____] weeks prior to the purchase or installation of a particular component, highlighted to show features such as materials of construction, dimensions, options, performance and efficiency. Data must include manufacturer's recommended installation instructions and procedures. Provide installation instructions that include assembly, alignment of components, piping connections, electrical connections, start-up procedures and checklists. Consider all recommendations in the written instructions of the manufacturer as mandatory requirements. Data must be adequate to demonstrate compliance with contract requirements.

3.2.2 Vibration Isolation

If vibration isolation is specified for a unit, vibration isolator literature must be included containing catalog cuts and certification that the isolation characteristics of the isolators provided meet the manufacturer's recommendations.

3.2.3 Posted Instructions

Provide posted instructions, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

3.2.4 Verification of Dimensions

Prior to beginning construction, the Contractor and the manufacturer's representative must become familiar with all details of the work, verify all dimensions in the field, verify clearances to enclosures, maintain manufacturer clearances to prevent recirculation, and advise the Contracting Officer of any discrepancy before performing any work. Before performing any work, the manufacturer representative must provide a letter stating the dimensions and the clearances to enclosures are within the recommendations of the equipment. Provide a letter including the date the site was visited, conformation of existing conditions, and any discrepancies found.

3.2.5 Demonstrations

Provide a schedule, at least [2][_____] weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.

3.2.6 Certificates

Where the system, components, or equipment are specified to comply with requirements of AGA, NFPA, ARI, ASHRAE, ASME, or UL, proof of such compliance must be provided. The label or listing of the specified agency must be acceptable evidence. In lieu of the label or listing, a written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of the specified agency may be submitted. When performance requirements of this project's drawings and specifications vary from standard ARI rating conditions, computer printouts, catalog, or other application data certified by ARI or a nationally recognized laboratory as described above must be included. If ARI does not have a current certification program that encompasses such application data, the manufacturer may self-certify that his application data complies with project performance requirements in accordance with the specified test standards.

[3.2.7 Start-Up Testing Report

Provide a start-up report documenting the start-up of the cooling tower. In the start-up report document cleaning and disinfection of the tower, passivation, filling of the tower, application of chemical treatment, compliance inspections, and start up the fan(s), basin heater, make-up water system, drainage system, and freeze protection devices.

]3.2.8 Operation and Maintenance Manuals

Provide [Six] [_____] complete copies of an operation manual in bound 216 by 279 mm 8 1/2 by 11 inch booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least [4][_____] weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features. [Six][_____] complete copies of maintenance manual in bound 216 by 279 8 1/2 by 11 inch booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts, and simplified wiring and control diagrams of the system as installed.

3.2.9 Connections to Existing Systems

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Furnish materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Furnish labor and tools for making actual connections to existing systems.

[3.3 RELATED FIELD TESTING

3.3.1 Test Plans

- a. Manufacturer's Test Plans: Within [120][_____] calendar days after contract award, submit the following plans:

- [(1) Packaged cooling tower - field acceptance test plan
][(2) Field-erected cooling tower - field acceptance test plan

Field acceptance test plans must be developed by the cooling tower manufacturer detailing recommended field test procedures for that particular type and size of equipment. Field acceptance test plans must be developed by the installing Contractor, or the equipment sales agency furnishing the equipment, must not be acceptable.

The Contracting Officer reviews and approves the field acceptance test plan for each of the listed equipment prior to commencement of field testing of the equipment. The approved field acceptance test plans must be the plan and procedures followed for the field acceptance tests of the cooling towers and subsequent test reporting.

] *****
NOTE: In the item below, Section 23 09 53.00 20
SPACE TEMPERATURE CONTROL SYSTEMS is for Navy
projects only.

- b. Coordinated testing: Indicate in each field acceptance test plan when work required by this section requires coordination with test work required by other specification sections. Furnish test procedures for the simultaneous or integrated testing of tower system controls which interlock and interface with controls for the equipment provided under [Section 23 09 53.00 20, SPACE TEMPERATURE CONTROL SYSTEMS][Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC][Section 23 09 23.01 LONWORKS DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS][Section 25 10 10 UTILITY MONITORING AND CONTROL SYSTEM (UMCS) FRONT END AND INTEGRATION][or][Section 23 09 23.02 BACNET DIRECT DIGITAL CONTROL FOR HVAC AND OTHER BUILDING CONTROL SYSTEMS].
- c. Prerequisite testing: Cooling towers for which performance testing is dependent upon the completion of the work covered by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must have that work completed as a prerequisite to testing work under this section. Indicate in each field acceptance test plan when such prerequisite work is required.
- d. Test procedure: Indicate in each field acceptance test plan each equipment manufacturers published installation, start-up, and field acceptance test procedures. Include in each test plan a detailed step-by-step procedure for testing automatic controls provided by the manufacturer.

Each test plan must include the required test reporting forms to be completed by the Contractor's testing representatives. Procedures must be structured to test the controls through all

modes of control to confirm that the controls are performing with the intended sequence of control.

Controllers must be verified to be properly calibrated and have the proper set point to provide stable control of their respective equipment.

- e. Performance variables: Each test plan must list performance variables that are required to be measured or tested as part of the field test.

Include in the listed variables performance requirements indicated on the equipment schedules on the design drawings. Tower manufacturer must furnish with each test procedure a description of acceptable results that have been verified.

Tower manufacturer must identify the acceptable limits or tolerances within which each tested performance variable must acceptably operate.

- f. Job specific: Each test plan must be job specific and must address the particular cooling towers and particular conditions which exist with this contract. Generic or general preprinted test procedures are not acceptable.
- g. Specialized components: Each test plan must include procedures for field testing and field adjusting specialized components, such as hot gas bypass control valves, or pressure valves.

]3.4 TESTING

- a. Each cooling tower system must be field acceptance tested[for thermal performance][, efficiency][, noise][, and vibration] in compliance with its approved field acceptance test plan and the resulting following field acceptance test report submitted for approval:

[(1) [Packaged cooling tower - field acceptance test report](#)

][(2) [Field-erected cooling tower - field acceptance test report](#)

-] b. Manufacturer's recommended testing: Conduct the manufacturer's recommend field testing in compliance with the approved test plan. Furnish a factory trained field representative authorized by and to represent the equipment manufacturer at the complete execution of the field acceptance testing.
- c. Operational test: Conduct a continuous 24 hour operational test for each item of equipment. Equipment shutdown before the test period is completed must result in the test period being started again and run for the required duration. For the duration of the test period, compile an operational log of each item of equipment. Log required entries every two hours. Use the test report forms for logging the operational variables.
- d. Notice of tests: Conduct the manufacturer's recommended tests and the operational tests; record the required data using the approved reporting forms. Notify the Contracting Officer in writing at least 15 calendar days prior to the testing. Within 30 calendar days after acceptable completion of testing, submit each test report for review and approval.

- e. Report forms: Type data entries and writing on the test report forms. Completed test report forms for each item of equipment must be reviewed, approved, and signed by the Contractor's test director. The manufacturer's field test representative must review, approve, and sign the report of the manufacturer's recommended test. Signatures must be accompanied by the person's name typed.
- f. Deficiency resolution: The test requirements acceptably met; deficiencies identified during the tests must be corrected in compliance with the manufacturer's recommendations and corrections retested in order to verify compliance.
- g. Towers with thermal performance not CTI certified to CTI Std-201 must have their thermal performance verified by field testing that meets the requirements of CTI ATC-105

-- End of Section --